

# Eduardo de la Rosa Flores

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## Professional Objective

Innovative Mechatronics Engineer with a strong foundation in bridging complex systems through software development and data-driven approaches. **Highly adaptable and driven by a continuous learning mindset**, I am eager to master new technologies to develop scalable, high-impact solutions across diverse engineering and technology sectors.

## Education

### Instituto Tecnológico y de Estudios Superiores de Monterrey

B.S. in Mechatronics Engineering

Monterrey, MX

Aug. 2020 – Jun. 2026

- **GPA: 90/100** | *DHAK Double Degree Program, scholarship awarded for academic excellence.*
- **Relevant Courses:** Firmware Testing, Communication Protocols, Industrial Automation, Robotic Software.
- **Extracurricular:** German Teacher & Study Abroad Mentor

### Esslingen University of Applied Sciences

B.Eng. in Automation Systems and Production Informatics

Esslingen, DE

Sep. 2024 – Feb. 2026

- **Relevant Courses:** AI and Machine Learning, Project Management and Robotics.

## Skills

**Languages:** Spanish (Native), English (C1), German (B2), French (A2)

**Soft Skills:** High Agency • Systems Thinking • Assertive and Efficient Communication • Adaptability • Leadership

**Technical Skills:**

*Software & Data Engineering:* Python, C/C++, Generative AI Integration, REST APIs & Full-stack Development

*Embedded Systems & Validation:* Microcontroller Architectures, RTOS, HIL Testing, Automotive Networks & Serial Protocols

*Robotics, Automation & DevOps:* Industrial Automation, PLC Programming, Embedded Linux, DevOps & Version Control

## Experience

### BMW Group

Munich, DE

**Bachelor's Thesis Student** *Quality Management R&D – LLM-Assisted Event Matching*

Sep. 2025 – Jan. 2026

- Centralized supplier-quality data in **Palantir Foundry** by combining 5 different databases, enabling seamless tracking of parts and suppliers before and after production launch (SOP).
- Developed an AI pipeline (**GPT-4.1 & PySpark**) to automatically identify if factory production issues had already appeared during early testing phases, achieving **94% accuracy**.
- Secured 100% data reliability through strict **JSON validation** at a cost under **\$0.002 per incident**, deploying interactive **Dashboards** for real-time team review.

**Data Engineering & AI Integration Intern** *Supply Chain Quality Management*

Mar. 2025 – Sep. 2025

- **Automated big-data ETL pipelines** in Python that integrated LLM analytics via **APIs**, processing hundreds of thousands of quality records to replace manual report reading with fast automated summaries.
- **Designed automated dashboards** to analyze supplier performance for a specific vehicle model, **utilized by a 15-engineer team** and scaled across multiple business units.
- **Built and deployed a cloud-based gamified web app**, integrating the Google Fit API for live team leaderboards.
- Completed a hands-on assembly line rotation on underbody components, bridging digital and physical manufacturing.

### Éxodo Al-Melek

Monterrey, MX

**Regional Subcoordinator**

Jul. 2021 – Jul. 2024

- Led a multidisciplinary team of 20 people to **plan, organize, and execute regional events** for 150+ participants.
- Adapted communication styles across multi-generational groups to build trust and ensure alignment.

## Technical Projects

**HIL Test Bench for Power Management in the Automotive Industry** – *Partner: International Motors Mexico*

- Engineered a 4-channel isolated power board (**12V/5A**) using solid state switching for automated ECU validation.
- Developed **ESP32** firmware parsing standard **VISA commands** to stream real-time **I2C (INA226)** telemetry.
- Designed a multi-threaded **Python/Tkinter GUI** to automate power-cycling sequences and monitor live current/voltage waveforms.

**Autonomous Mobile Robotics** – *Partner: John Deere – Advanced Control*

- Designed modular navigation firmware with decoupled architectures for wheel control, enhancing system scalability.
- Identified system with **ARX modeling** to tune **closed-loop** speed and steering controllers, minimizing trajectory deviation.
- Fused **I2C (MPU6050)** and **UART** sensor data via filtering, mitigating dead-reckoning drift during autonomous navigation.

**Smart Tire Inflation Platform** – *Partner: John Deere – Advanced Embedded Systems*

- Engineered an automated pressure system using an **STM32H7** microcontroller communicating over a local **CAN network**.
- Implemented a **closed-loop PI controller** to regulate a 12V compressor and solenoid valve based on terrain profiles.
- Deployed a **NodeMCU Wi-Fi gateway** to stream real-time sensor telemetry to an **Arduino IoT Cloud** dashboard.

**Embedded Audio System** – *Partner: Intel – RTOS & Multithreading*

- Engineered a multithreaded **FreeRTOS** architecture for concurrent UI management and real-time audio task scheduling.
- Optimized IPC via queues and semaphores to ensure race-free, zero-jitter **DMA audio playback**.
- Developed a custom serial communication protocol for strict metadata and state synchronization with an **Embedded Linux** server.